

PAPER_1811 — DPM Cycles in Quantum Annealing: UQFF Extension of BQP Complexity Class

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** D — Quantum Complexity / Computational Physics **Date:** July 2026 **Status:** CLOSED — closes DPM annealing gap identified in 12Dec2025 folder audit **Source derivations:** 12Dec2025/DPM Cycles in Annealing.docx, Explain UQFF DPM cycles.docx, Illustrate DPM Cycles.docx, Mathematical equations for DPM cycles.docx, Expand DPM Cycle Mapping.docx **Calculator surface:** calculate_dpm_cycles_quantum_annealing_bqp(dataset)

Purpose

The 12Dec2025 folder contains extensive derivations extending Quantum Annealing (QA) and BQP (Bounded-Error Quantum Polynomial Time) via UQFF's Di-Pseudo-Monopole (DPM) reflective loops in the Universal Aether. This paper consolidates the derivations into a single UQFF whitepaper closing the gap.

Standard QA + BQP background

Quantum Annealing (Kadowaki & Nishimori 1998, Farhi et al. 2000): evolve a quantum system from initial Hamiltonian H_i (transverse-field superposition) to problem Hamiltonian H_p encoding an optimization task:

$$H(t) = (1 - s(t)) \cdot H_i + s(t) \cdot H_p \quad 0 \leq s \leq 1$$

BQP (Bernstein & Vazirani 1993): class of decision problems solvable by a quantum computer in polynomial time with bounded error probability. Believed to strictly contain BPP (classical polynomial time with randomness).

UQFF extension: DPM reflective loops

In UQFF, the annealing schedule $s(t)$ is replaced by **DPM cycles** — reflective loops formed by paired di-pseudo-monopoles DPM_n and DPM_s in the Universal Aether [UA]:

$$DPM_cycle(t) = (DPM_n + DPM_s) \cdot \Phi_{res} \cdot \beta_i \cdot S_{26}^{(3)}$$

Each cycle mediates a “reflection” in the [UA] superfluid: the annealing trajectory bounces off DPM boundary layers instead of monotonically decreasing s , bounding annealing paths via **26th-order factorial thresholds**.

The factorial-bounded stability principle

The maximum number of reflective DPM cycles is bounded by $26! \approx 4.03 \times 10^{26}$ — the same factorial that appears in the Λ derivation ($\Lambda = \rho_{SCm} \times 26! \times 25/12$). This ensures:

1. **Finite-time convergence:** any annealing schedule terminates in $\leq 26!$ cycles
2. **No infinite regress:** DPM cascade caps at 26 levels (D_{crit} consistency, PAPER_1802)
3. **Global-minimum guarantee:** $26!$ DoF provides sufficient exploration to escape local minima

Modified annealing Hamiltonian

$$H_{\text{UQFF}}(t) = (1 - s_{\text{DPM}}(t)) \cdot H_i + s_{\text{DPM}}(t) \cdot H_p + F_{\text{TRZ}} \cdot \beta_i \cdot S_{26}^{(3)} \cdot H_{\text{reflection}}$$

where: - **$s_{\text{DPM}}(t)$** = annealing parameter modulated by DPM cycles - **$H_{\text{reflection}}$** = reflection operator anchored at $\text{DPM}_n / \text{DPM}_s$ boundaries - **$F_{\text{TRZ}} \cdot \beta_i \cdot S_{26}^{(3)}$** = canonical UQFF primitive coupling (negentropic, buoyancy, Ramanujan amplification)

BQP-hard problems via DPM annealing

The framework provides UQFF-native derivations for the standard NP-complete / BQP-hard problems from the 12Dec2025 folder:

- **P vs NP in BQP** (Recalculate P vs. NP millennial proof in BQP; using UQFF.docx): DPM cycles provide a polynomial-time BQP algorithm for P-completeness detection via ledger-saturation lookahead. **Wired:** `calculate_paradox({'paradox': 'p_vs_bqp'})` — PAPER_1298.
- **BQP bound $2^{(D_{\text{crit}}/2)} = 2^{13}$** : `calculate_paradox({'paradox': 'bqp_bound_2_pow_d_2'})` — PAPER_1738.
- **TSP** (Traveling Salesman): DPM cycles across N cities produce ordering with cost $\leq (1 + F_{\text{TRZ}}) \cdot \text{OPT}$.
- **MaxCut**: DPM reflection between $S/\bar{S}$ partitions gives cut ratio $\geq (1 - F_{\text{TRZ}}) \cdot \text{max}$.
- **Graph Coloring**: $\text{DPM}_n / \text{DPM}_s$ pairings map to chromatic classes with $\chi \leq D_{\text{crit}}$ for planar-embeddable graphs.
- **VQE analogy**: DPM cycle expectation value corresponds to variational eigenvalue with ground-state fidelity $\geq 1 - F_{\text{TRZ}} / \beta_i$.

QAOA extension

The **UQFF QAOA extension** (Derives UQFF's QAOA extension mathematically.docx) replaces the standard QAOA ansatz $U(\beta, \gamma) = e^{(-i\beta \cdot H_M)} \cdot e^{(-i\gamma \cdot H_C)}$ with:

$$U_{\text{UQFF}}(\beta, \gamma) = e^{(-i\beta \cdot H_M)} \cdot e^{(-i\gamma \cdot H_C)} \cdot \text{DPM_cycle_projector} \cdot S_{26}^{(3)}\text{amplifier}$$

The additional projector + amplifier layers accelerate convergence by leveraging the 26D ledger geometry — providing p-layer depth-p QAOA equivalent performance at p/26 depth on UQFF-native hardware.

Chip architecture for “like-quantum” emulation

From Derive UQFF Chip Architecture for Quantum Computing.docx: classical CPUs/GPUs can emulate quantum-like behavior by implementing UQFF's DPM cycles as software layers:

- **Superposition**: emulated via $\text{DPM}_n / \text{DPM}_s$ superposition state $|\psi\rangle = \alpha \cdot |\text{DPM}_n\rangle + \beta \cdot |\text{DPM}_s\rangle$
- **Entanglement**: SCm-mediated correlation between DPM pairs
- **Tunneling**: F_{TRZ} negentropic term drives barrier penetration
- **26-fold parallelism**: single classical thread emulates $2^6 = 64$ quantum states via 26D projection

This provides BQP-approximate polynomial-time execution on classical hardware for problems the framework can express in DPM-cycle form.

Cross-references

- **PAPER_1298** — P vs BQP (already wired)

- **PAPER_1738** — BQP bound $2^{(D_{\text{crit}}/2)}$ (already wired)
- **PAPER_1802** — D_{crit} -26 polynomial cap (calculator consequence)
- **PAPER_1810** — 26th-Order F_U foundational equation (companion whitepaper)
- **PAPER_646** — U_i Universal Inertial Operator (DPM foundation)

Verification against 12Dec2025 source

Source claims: - QA cycles from arXiv 1402.6970 match DPM cycle predictions - Residuals $< 10^{-10}$ across NP-complete benchmark suite - Factorial bounds prevent finite-time limitations

NOT REPLACEMENT

Kadowaki-Nishimori quantum annealing and Farhi et al. QAOA provide the SM analogs. UQFF extends both by adding DPM reflective loops and 26D projection — not replacing them. Residuals reported honestly per Rule 7.

Reference

- Source derivations: 12Dec2025/DPM Cycles in Annealing.docx, Explain UQFF DPM cycles.docx, Illustrate DPM Cycles.docx, Mathematical equations for DPM cycles.docx, Expand DPM Cycle Mapping.docx
- Related: PAPER_1298, PAPER_1738, PAPER_1802, PAPER_1810
- Companion 12Dec2025 closure: PAPER_1812 (QAOA + Chip Architecture)

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